INSTRUMENT CLUSTER (Continued)

function Indicator Lamp (MIL) is normally controlled by PCI data bus messages from the PCM; however, if the EMIC loses PCI data bus communications, the EMIC circuitry will automatically turn the MIL on and display the message “no BuS” in the odometer VFD unit until PCI data bus communication is restored. The EMIC uses PCI data bus messages from the PCM, Airbag Control Module (ACM), and the Sentry Key Immobilizer Module (SKIM) to control all of the remaining indicators.

The various EMIC indicators are controlled by different strategies; some receive fused ignition switch output from the EMIC circuitry and have a switched ground, others are grounded through the EMIC circuitry and have a switched battery feed. However, all indicators except those for the antilock brake system, turn signals, and fog lamps are completely controlled by the EMIC microprocessor based upon various hard wired and electronic message inputs. The cruise, four-wheel drive and both axle lock indicators are one-step dimmable based upon an input to the EMIC from the headlamp switch circuitry. When the exterior lamps are off, these indicators are illuminated at a fixed maximum intensity; and, when the exterior lamps are on, these indicators are dimmed by the EMIC to a fixed lower intensity. All remaining indicators are illuminated at a fixed intensity, which is not affected by the status of the exterior lighting or the selected illumination intensity of the EMIC general illumination lamps.

In addition, certain indicators in this instrument cluster are automatically configured or self-configured. This feature allows the configurable indicators to be enabled by the EMIC circuitry for compatibility with certain optional equipment. The airbag indicator, cruise indicator, and the Sentry Key Immobilizer System (SKIS) indicator are automatically configured. The automatically configured or self-configured indicators remain latent in each EMIC at all times and will be active only when the EMIC receives the appropriate PCI message inputs for that optional system or equipment. Once a configurable indicator is enabled by the EMIC, it is learned and stored in cluster memory for the remainder of the cluster life.

The hard wired indicator inputs may be diagnosed using conventional diagnostic methods. However, the EMIC circuitry and PCI bus message controlled indicators are diagnosed using the EMIC self-diagnostic actuator test. (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING). Proper testing of the PCI data bus and the electronic message inputs to the EMIC that control an indicator requires the use of a DRBIII® scan tool. Refer to the appropriate diagnostic information. Specific details of the operation for each indicator may be found elsewhere in this service information.

CLUSTER ILLUMINATION

The EMIC has several illumination lamps that are illuminated when the exterior lighting is turned on with the headlamp switch circuitry of the left multi-function switch. The illumination intensity of these lamps is adjusted when the interior lamps control ring on the left multi-function switch control stalk is rotated (downward to dim, upward to brighten). In response to that input, an analog/digital (A/D) converter in the EMIC converts the analog panel lamps dimmer resistor multiplexed input from the left multi-function switch into a digital dimming level message and a 12-volt Pulse-Width Modulated (PWM) output. The EMIC uses the PWM output to power the cluster illumination lamps and the VFD unit on the EMIC circuit board, then provides a synchronized PWM output on the hard wired fused panel lamp feed output circuit to control and synchronize the illumination intensity of other incandescent illumination lamps in the vehicle. The cluster illumination lamps are grounded at all times.

The EMIC also sends electronic dimming level messages over the PCI data bus to other electronic modules in the vehicle to control and synchronize the illumination intensity of their VFD units with that of the EMIC VFD unit. In addition, the interior lamps control ring on the left multi-function switch control stalk has a Parade Mode position to provide a parade mode. The EMIC monitors the request for this mode from the left multi-function switch, then sends an electronic dimming level message over the PCI data bus to illuminate all VFD units in the vehicle at full (daytime) intensity for easier visibility when driving in daylight with the exterior lighting turned on.

The hard wired left multi-function switch cluster illumination inputs and the EMIC fused panel lamp feed output may be diagnosed using conventional diagnostic methods. However, proper testing of the PWM control of the EMIC and the electronic dimming level messages sent by the EMIC over the PCI data bus requires the use of a DRBIII® scan tool. Refer to the appropriate diagnostic information.

AUDIO SYSTEM CABIN EQUALIZATION

Each time the EMIC receives an electronic cabin equalization request message from the radio over the PCI data bus, it provides an electronic response to the radio containing the appropriate equalization curve information. Because there are numerous optional radios which are common to many platforms and available with various speaker architectures, each radio contains a Digital Signal Processing (DSP) microprocessor chip. This DSP chip uses the equalization curve information to optimize the radio’s sound output characteristics for the unique cabin and